Learning for Data Science

Course 10: Practical Hypothesis Testing and Experimental Design

Glen Berseth

Université de Montréal and Mila Québec AI Institute

October 6, 2026



Outline

Introduction: Inference

Bootstrapping: Yertle's Turtle Tower

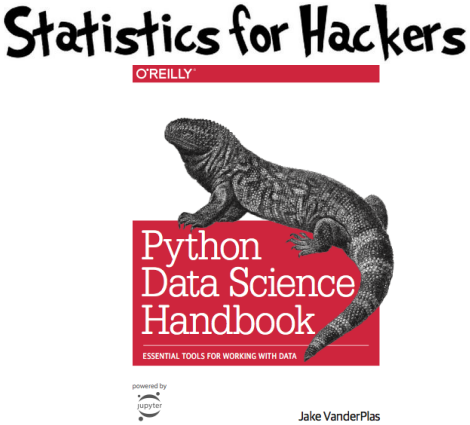
Cross Validation

Summary: Four Recipes for Hacking Statistics

Introduction: Inference

- Meta-Algorithms: Evaluation of processes people use to learn from data
- Science: How to go from the finite to the general?
- "It is easy to misinterpret with statistics, but hard to tell the truth without them."

Special Thanks





“Why don’t you just . . .”

< One popular alternative . . . >

```
from statsmodels.stats.weightstats import ttest_ind
t, p, dof = ttest_ind(group1, group2,
                      alternative='larger',
                      usevar='unequal')
print(p)   # 0.186
```

“Why don’t you just . . .”

< One popular alternative . . . >

```
from statsmodels.stats.weightstats import ttest_ind
t, p, dof = ttest_ind(group1, group2,
                      alternative='larger',
                      usevar='unequal')
print(p)   # 0.186
```

. . . But what *question* is this answering?

Stepping Back...

The deep meaning lies in the *sampling distribution*:
Central Limit Theorem:

$$X_1, \ldots, X_n \text{ i.i.d.}, \quad \mathbb{E}[X_i] = \mu, \quad \mathbb{E}[(X_i - \mu)^2] = \sigma^2$$

$$\frac{\sum_{i=1}^{n} X_i - n\mu}{\sqrt{n}} \to \mathcal{N}(0, \sigma^2)$$

$$t = \underbrace{\frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}}$$

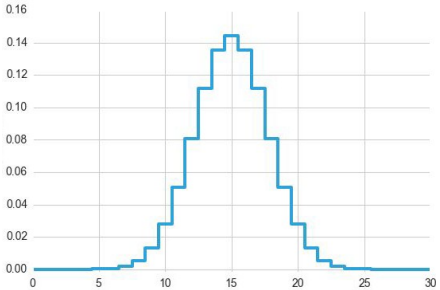
"Look like" Gaussian variables

Stepping Back...

The deep meaning lies in the *sampling distribution*:

$$p(t;\nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

Same principle as the coin example:



0.8 %

Use a Sampling Method Instead

The Problem:
Unlike coin flipping, there is no *generative model* . . .

Use a Sampling Method Instead

The Problem:
Unlike coin flipping, there is no *generative model* . . .
Solution: Shuffling

Shuffling

⋆		×	
84	72	81	69
57	46	74	61
63	76	56	87
99	91	69	65
		66	44
		62	69

Idea:
Simulate the distribution by *shuffling* the labels repeatedly and computing the desired statistic.

Motivation:
If the labels really don't matter, then switching them shouldn't change the result!

Shuffling

⋆		×	
84	72	81	69
57	46	74	61
63	76	56	87
99	91	69	65
		66	44
		62	69

1. Shuffle Labels
2. Rearrange
3. Compute means

Shuffling

⋆		×	
84	72	81	69
57	46	74	61
63	76	56	87
99	91	69	65
		66	44
		62	69

1. **Shuffle Labels**
2. Rearrange
3. Compute means

Shuffling

$\star$		$\times$	
84	81	72	69
61	69	74	57
65	76	56	87
99	44	46	63
		66	91
		62	69

1. Shuffle Labels
2. **Rearrange**
3. Compute means

Shuffling

⋆		×	
84	81	72	69
61	69	74	57
65	76	56	87
99	44	46	63
		66	91
		62	69

- ⋆ mean: 72.4
- × mean: 67.6
- Difference: 4.8

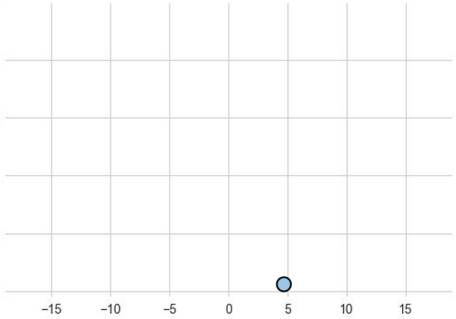
1. Shuffle Labels
2. Rearrange
3. **Compute means**

Shuffling

⋆		×	
84	81	72	69
61	69	74	57
65	76	56	87
99	44	46	63
		66	91
		62	69

- ⋆ mean: 72.4
- × mean: 67.6
- Difference: 4.8

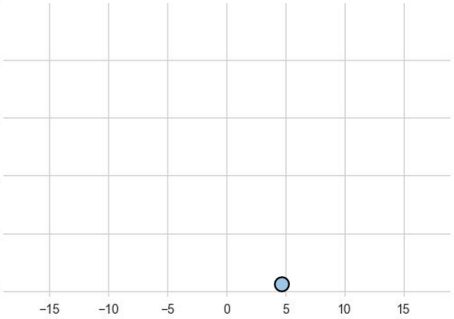
1. Shuffle Labels
2. Rearrange
3. **Compute means**



Shuffling

⋆		×	
84	81	72	69
61	69	74	57
65	76	56	87
99	44	46	63
		66	91
		62	69

1. **Shuffle Labels**
2. Rearrange
3. Compute means

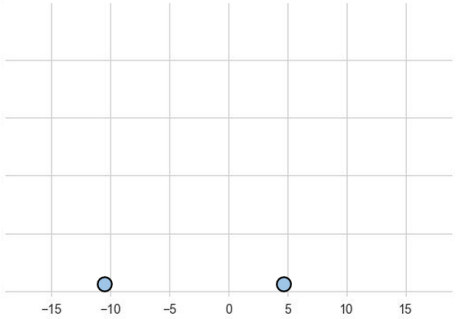


Shuffling

⋆		×	
84	56	72	69
61	63	74	57
65	66	81	87
62	44	46	69
		76	91
		99	69

- ⋆ mean: 62.6
- × mean: 74.1
- Difference: -11.6

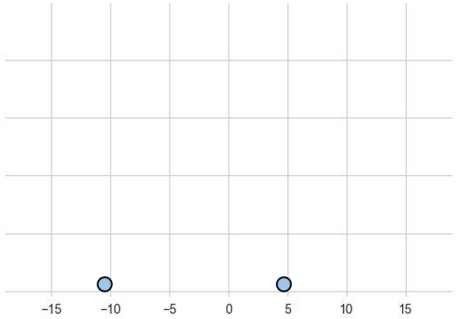
1. Shuffle Labels
2. Rearrange
3. **Compute means**



Shuffling

⋆		×	
84	56	72	69
61	63	74	57
65	66	81	87
62	44	46	69
		76	91
		99	69

1. **Shuffle Labels**
2. Rearrange
3. Compute means

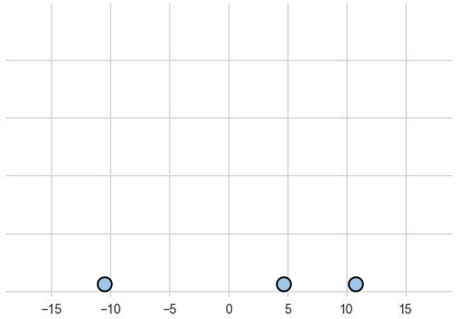


Shuffling

⋆		×	
74	56	72	69
61	63	84	57
87	76	81	65
91	99	46	69
		66	62
		44	69

- ⋆ mean: 75.9
- × mean: 65.3
- Difference: 10.6

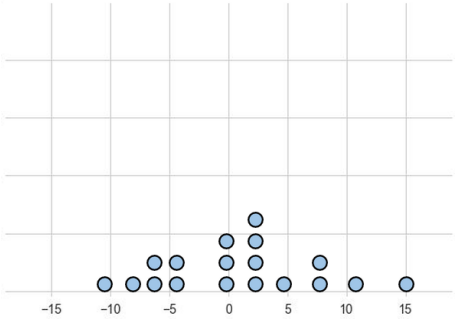
1. Shuffle Labels
2. Rearrange
3. **Compute means**



Shuffling

⋆		×	
84	56	72	69
61	63	74	57
65	66	81	87
62	44	46	69
		76	91
		99	69

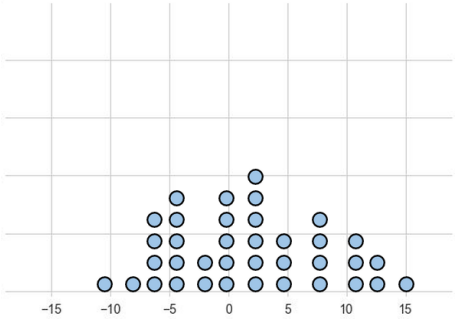
1. Shuffle Labels
2. Rearrange
3. Compute means



Shuffling

⋆		×	
84	81	69	69
61	69	87	74
65	76	56	57
99	44	46	63
		66	91
		62	72

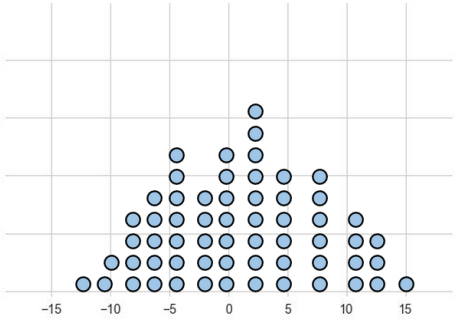
1. Shuffle Labels
2. Rearrange
3. Compute means



Shuffling

⋆		×	
74	62	72	57
61	63	84	69
87	81	76	65
91	99	46	69
		66	56
		44	69

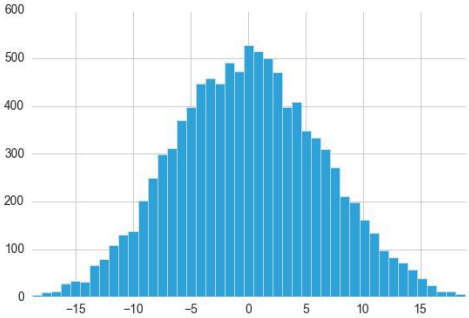
1. Shuffle Labels
2. Rearrange
3. Compute means



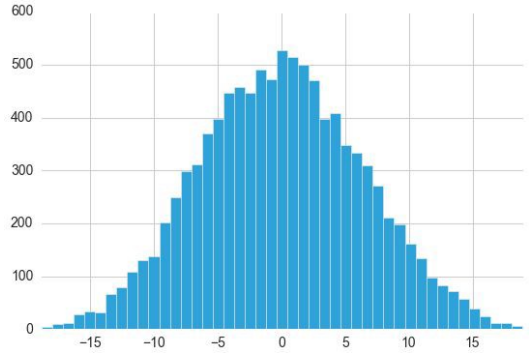
Shuffling

⋆		×	
84	81	72	69
61	69	74	57
65	76	56	87
99	44	46	63
		66	91
		62	69

1. Shuffle Labels
2. Rearrange
3. Compute means



Shuffling



score difference

16 %

$$\frac{N_{>6.6}}{N_{tot}} = \frac{1608}{10000} = 0.16$$

“A difference of 6.6 is not significant at $p = 0.05$.”



That day, all the Sneetches forgot about stars
And whether they had one, or not, upon thars.

Notes on Shuffling

- To measure a statistical significance between **two** groups.

Notes on Shuffling

- To measure a statistical significance between **two** groups.
- Works when the *Null Hypothesis* assumes the two groups are equivalent.

Notes on Shuffling

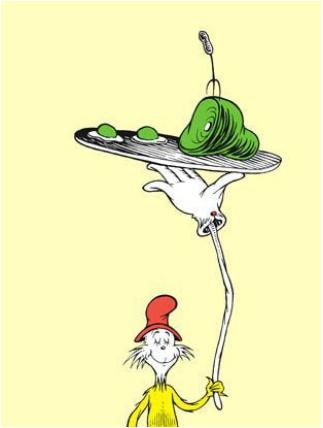
- To measure a statistical significance between **two** groups.
- Works when the *Null Hypothesis* assumes the two groups are equivalent.
- Like all methods, it will only work if your samples are representative – always be careful about selection biases!

Notes on Shuffling

- To measure a statistical significance between **two** groups.
- Works when the *Null Hypothesis* assumes the two groups are equivalent.
- Like all methods, it will only work if your samples are representative – always be careful about selection biases!
- Needs care for non-independent trials.
- Discussion in Simon's *Resampling: The New Statistics*.

Four Recipes for Hacking Statistics

1. Direct Simulation ✓
2. Shuffling ✓
3. Bootstrapping
4. Cross Validation



Bootstrapping: Yertle's Turtle Tower

On the far-away island of
Sala-ma-Sond,
Yertle the Turtle
was king of the pond. . .



How High can Yertle stack his turtles?

Observe 20 of Yertle's turtle towers . . .

# of turtles	48	24	32	61	51	12	32	18	19	24
	21	41	29	21	25	23	42	18	23	13

- What is the mean of the number of turtles in Yertle's stack?
- What is the uncertainty on this estimate?



Classic Method

Sample Mean:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i = 28.9$$

Standard Error of the Mean:

$$\sigma_{\bar{x}} = \frac{1}{\sqrt{N}}\sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2} = 3.0$$

What assumptions go into these formulae?

Can sampling be used instead?

Bootstrap Resampling

Problem:
As before, there is no *generating model* . . .

Bootstrap Resampling

Problem:
As before, there is no *generating model* . . .
Solution: Bootstrap Resampling

Bootstrap Resampling

48	24	51	12
21	41	25	23
32	61	19	24
29	21	23	13
32	18	42	18

Idea:
Simulate the distribution by *drawing samples with replacement.*

Motivation:
The data estimates its own distribution – random samples are drawn from this distribution.

Bootstrap Resampling

48	24	51	12
21	41	25	23
32	61	19	24
29	21	23	13
32	18	42	18

Idea:
Simulate the distribution by *drawing samples with replacement.*

Motivation:
The data estimates its own distribution – random samples are drawn from this distribution.

Bootstrap Resampling

48	24	51	12
21	41	25	23
32	61	19	24
29	21	23	13
32	18	42	18

Idea:
Simulate the distribution by *drawing samples with replacement.*

Motivation:
The data estimates its own distribution – random samples are drawn from this distribution.

21

Bootstrap Resampling

48	24	51	12
21	41	25	23
32	61	**19**	24
29	21	23	13
32	18	42	18

Idea:
Simulate the distribution by *drawing samples with replacement.*

Motivation:
The data estimates its own distribution – random samples are drawn from this distribution.

21	**19**

Bootstrap Resampling

48	24	51	12
21	41	**25**	23
32	61	19	24
29	21	23	13
32	18	42	18

Idea:
Simulate the distribution by *drawing samples with replacement.*

Motivation:
The data estimates its own distribution – random samples are drawn from this distribution.

21	19	**25**

Bootstrap Resampling

48	**24**	51	12
21	41	25	23
32	61	19	24
29	21	23	13
32	18	42	18

Idea:
Simulate the distribution by *drawing samples with replacement.*

Motivation:
The data estimates its own distribution – random samples are drawn from this distribution.

21	19	25	**24**

Bootstrap Resampling

48	24	51	12
21	41	25	23
32	61	19	24
29	21	**23**	13
32	18	42	18

Idea:
Simulate the distribution by *drawing samples with replacement.*

Motivation:
The data estimates its own distribution – random samples are drawn from this distribution.

21	19	25	24	**23**

Bootstrap Resampling

48	24	51	12
21	41	25	23
32	61	**19**	24
29	21	23	13
32	18	42	18

Idea:
Simulate the distribution by *drawing samples with replacement.*

Motivation:
The data estimates its own distribution – random samples are drawn from this distribution.

21	19	25	24	23	**19**

Bootstrap Resampling

48	24	51	12
21	**41**	25	23
32	61	19	24
29	21	23	13
32	18	42	18

Idea:
Simulate the distribution by *drawing samples with replacement.*

Motivation:
The data estimates its own distribution – random samples are drawn from this distribution.

21	19	25	24	23	19	41	23	41	18
61	12	42	42	42	19	18	61	29	**41**

Bootstrap Resampling

48	24	51	12
21	41	25	23
32	61	19	24
29	21	23	13
32	18	42	18

Idea:
Simulate the distribution by *drawing samples with replacement.*

Motivation:
The data estimates its own distribution – random samples are drawn from this distribution.

21	19	25	24	23	19	41	23	41	18
61	12	42	42	42	19	18	61	29	41

$\rightarrow 31.05$

Bootstrap Resampling

48	24	51	12
21	41	25	23
32	61	19	24
29	21	23	13
32	18	42	18

Idea:
Simulate the distribution by *drawing samples with replacement.*

Motivation:
The data estimates its own distribution – random samples are drawn from this distribution.

21	19	25	24	23	19	41	23	41	18
61	12	42	42	42	19	18	61	29	41

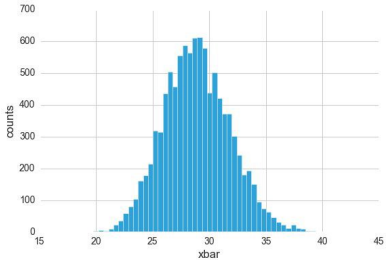
$\rightarrow$ 31.05

Repeat this several thousand times . . .

Recovers The Analytic Estimate!

```
for i in range(10000):
    sample = N[randint(20, size=20)]
    xbar[i] = mean(sample)
mean(xbar), std(xbar)
# (28.9, 2.9)
```

Height $= 29 \pm 3$ turtles





Notes on Bootstrapping

- Use to estimate statistics of your data (mean, std, . . .)

Notes on Bootstrapping

- Use to estimate statistics of your data (mean, std, ...)
- Well-studied and rests on solid theoretical grounds.

Notes on Bootstrapping

- Use to estimate statistics of your data (mean, std, ...)
- Well-studied and rests on solid theoretical grounds.
- Bootstrapping often **doesn't work** well for rank-based statistics (e.g. maximum value)

Notes on Bootstrapping

- Use to estimate statistics of your data (mean, std, ...)
- Well-studied and rests on solid theoretical grounds.
- Bootstrapping often **doesn't work** well for rank-based statistics (e.g. maximum value)
- Works poorly with very few samples ($N > 20$ is a good rule of thumb)

Notes on Bootstrapping

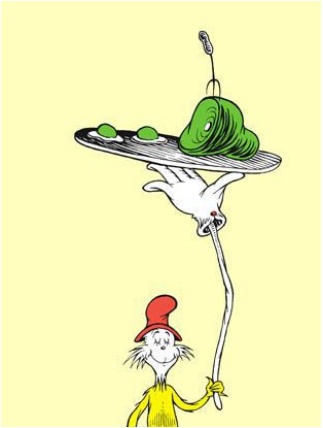
- Use to estimate statistics of your data (mean, std, ...)
- Well-studied and rests on solid theoretical grounds.
- Bootstrapping often **doesn't work** well for rank-based statistics (e.g. maximum value)
- Works poorly with very few samples ($N > 20$ is a good rule of thumb)
- As always, be careful about selection biases & non-independent data!

Notes on Bootstrapping

- Use to estimate statistics of your data (mean, std, ...)
- Well-studied and rests on solid theoretical grounds.
- Bootstrapping often **doesn't work** well for rank-based statistics (e.g. maximum value)
- Works poorly with very few samples ($N > 20$ is a good rule of thumb)
- As always, be careful about selection biases & non-independent data!
- More slides on bootstrapping

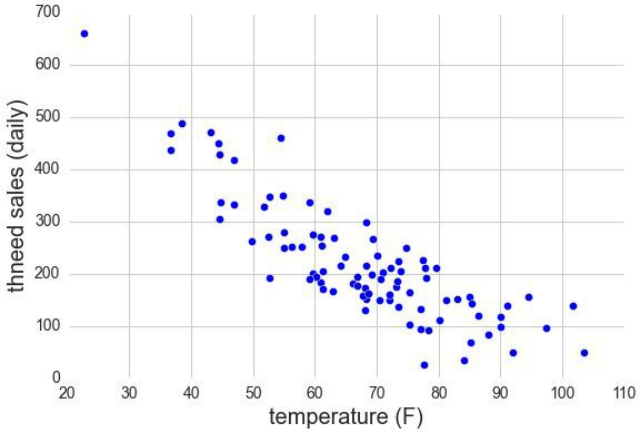
Four Recipes for Hacking Statistics

1. Direct Simulation ✓
2. Shuffling ✓
3. Bootstrapping ✓
4. Cross Validation

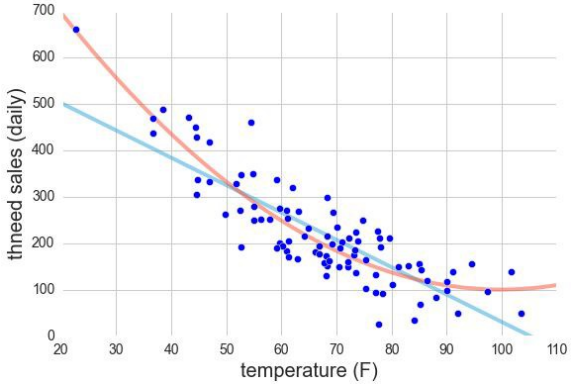


Cross Validation

- Sales seem to show a trend with temperature . . .



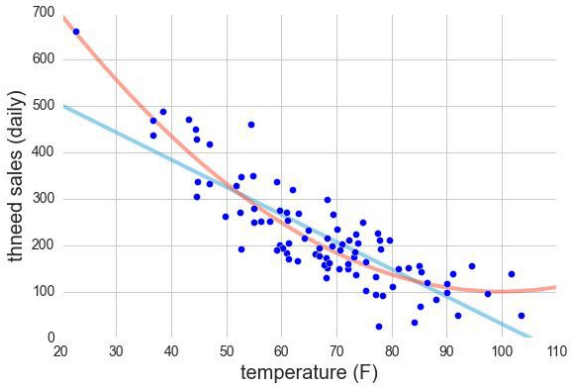
But which model is a better fit?



$y = a + bx$

$y = a + bx + cx^2$

But which model is a better fit?



$y = a + bx$

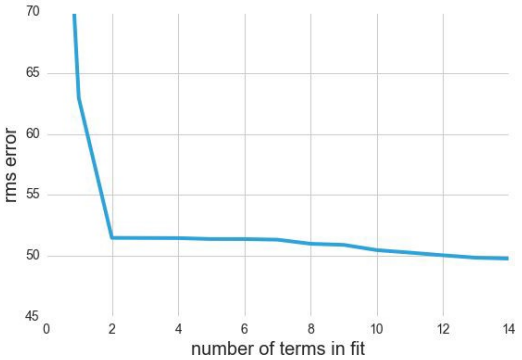
$y = a + bx + cx^2$

Can the models be judged by root-mean-square error?

RMS error $= 63.0$

RMS error $= 51.5$

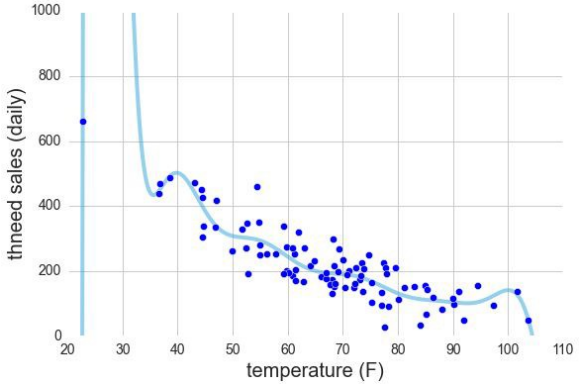
In general, more complex models will *always* have a lower RMS error



$$y = a + bx$$
$$y = a + bx + cx^2$$
$$y = a + bx + cx^2 + dx^3$$
$$y = a + bx + cx^2 + dx^3 + ex^4$$
$$y = a + \cdots$$

RMS error does not tell the whole story

$$y = a + bx + cx^2 + dx^3 + ex^4 + fx^5 + \cdots + nx^{14}$$



Not to worry: Statistics has figured this out



Classic Method

Difference in Mean Squared Error follows chi-square distribution:

$$p(x;\nu) = \frac{1}{2^{\nu/2}\,\Gamma\left(\frac{\nu}{2}\right)}\, x^{\frac{\nu}{2}-1} e^{-\frac{x}{2}}$$

Classic Method

Difference in Mean Squared Error follows chi-square distribution:

$$p(x;\nu) = \frac{1}{2^{\nu/2}\,\Gamma\left(\frac{\nu}{2}\right)}\, x^{\frac{\nu}{2}-1} e^{-\frac{x}{2}}$$

Can estimate degrees of freedom easily because the models are *nested* . . .

$$\nu \approx \nu_2 - \nu_1, \qquad \nu_2 \approx (N - d_2), \qquad \nu_1 \approx (N - d_1)$$

Classic Method

Difference in Mean Squared Error follows chi-square distribution:

$$p(x;\nu) = \frac{1}{2^{\nu/2}\,\Gamma\left(\frac{\nu}{2}\right)}\, x^{\frac{\nu}{2}-1} e^{-\frac{x}{2}}$$

Can estimate degrees of freedom easily because the models are *nested* . . .

$$\nu \approx \nu_2 - \nu_1, \qquad \nu_2 \approx (N - d_2), \qquad \nu_1 \approx (N - d_1)$$

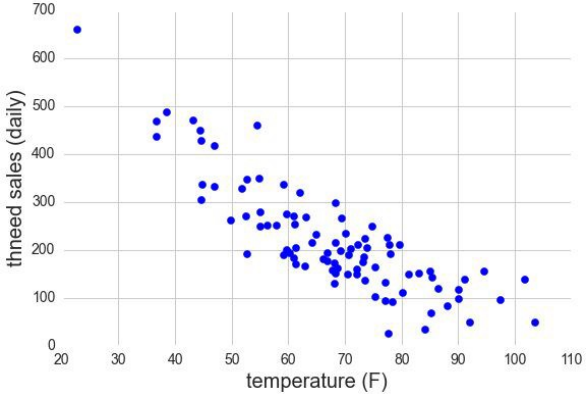
Plug in the numbers . . .

Classic Method

Wait... what question was this trying to answer again?

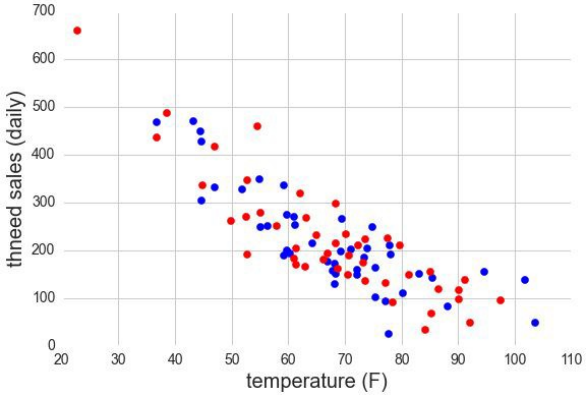
Cross-Validation

Another Approach: Cross Validation



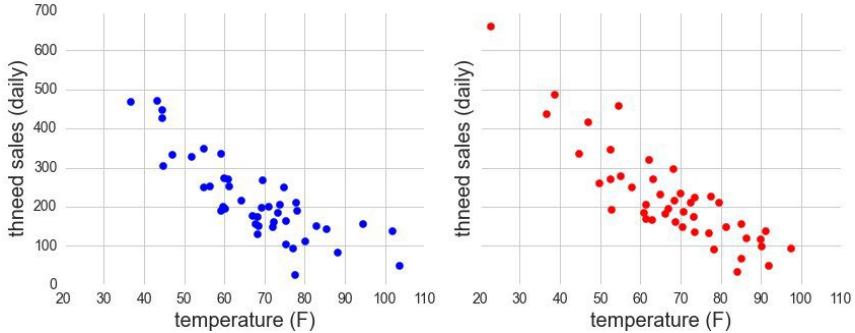
Cross-Validation

1. Randomly Split data

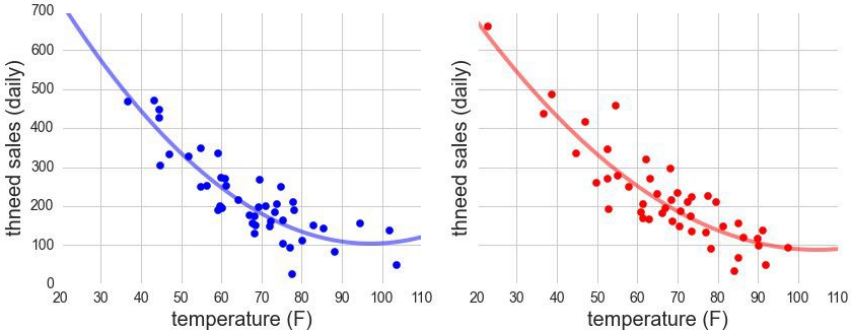


Cross-Validation

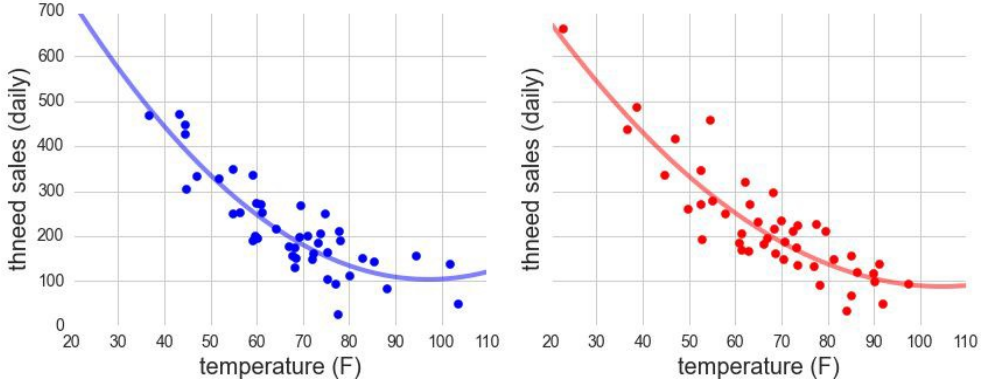
1. Randomly Split data



2. Find the best model for each subset

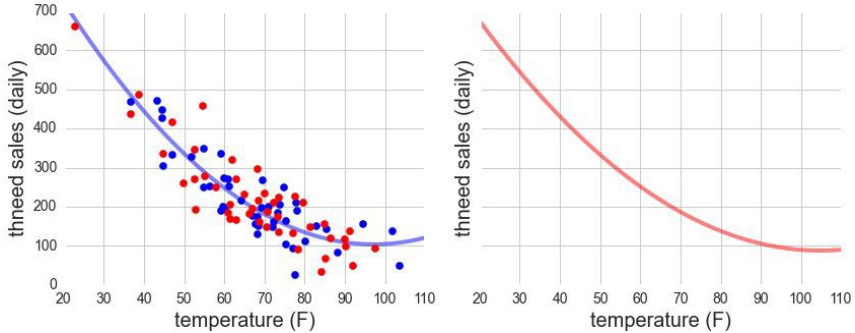


3. Compare models across subsets



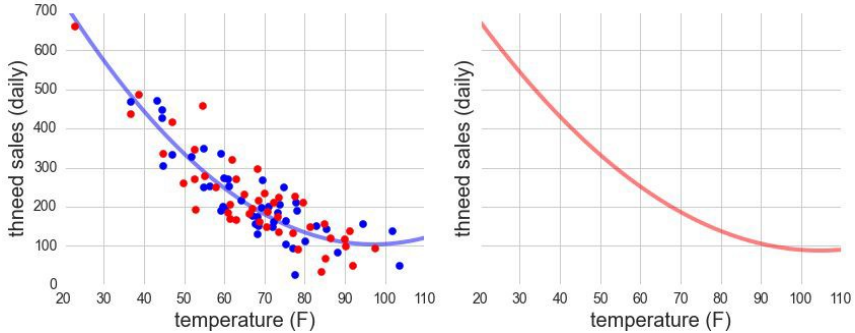
Cross-Validation

3. Compare models across subsets



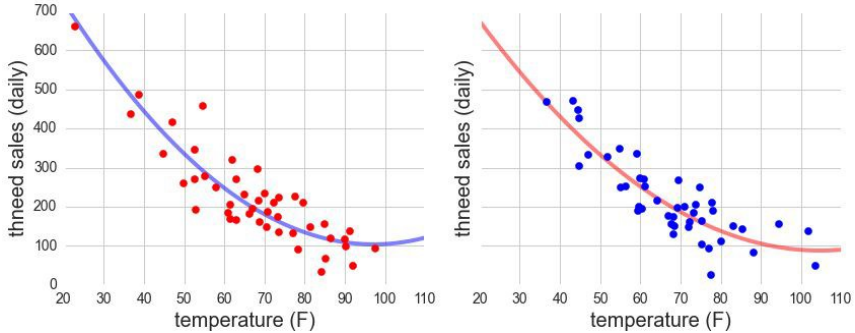
Cross-Validation

3. Compare models across subsets



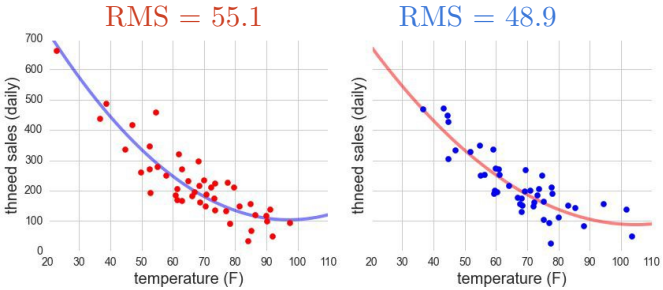
Cross-Validation

3. Compare models across subsets



Cross-Validation

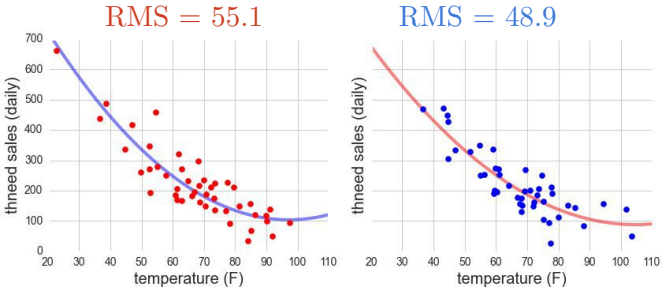
4. Compute RMS error for each



RMS estimate = 52.1

Cross-Validation

4. Compute RMS error for each

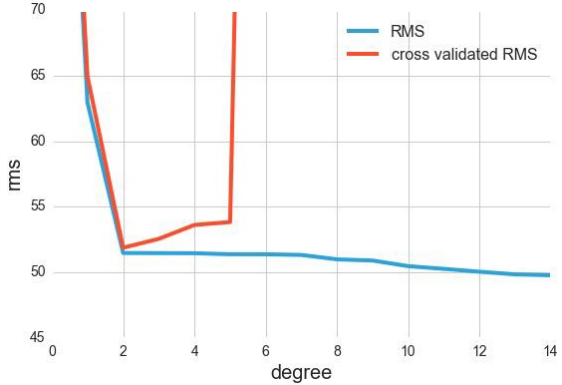


RMS estimate = 52.1

Repeat for as long as you have patience . . .

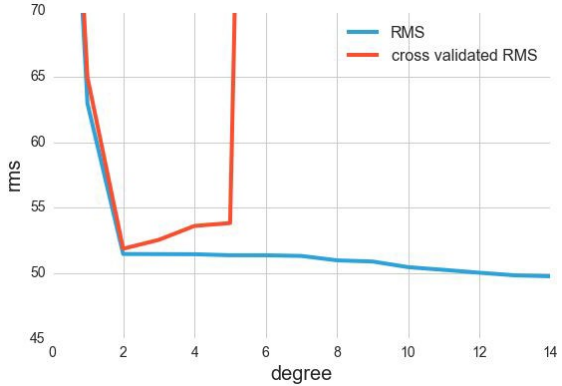
Cross-Validation

5. Compare cross-validated RMS for models:



Cross-Validation

5. Compare cross-validated RMS for models:



Best model minimizes the cross-validated error.

Notes on Cross-Validation

- This was **"2-fold" cross-validation**; other CV schemes exist & may perform better for your data (see e.g. scikit-learn docs)

Notes on Cross-Validation

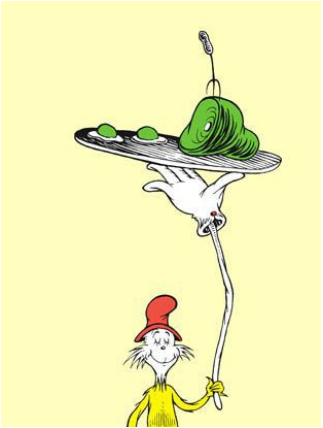
- This was **"2-fold" cross-validation**; other CV schemes exist & may perform better for your data (see e.g. scikit-learn docs)
- Cross-validation is the go-to method for model evaluation in **machine learning**, as statistics of the models are often not known in the classical sense.

Notes on Cross-Validation

- This was **"2-fold" cross-validation**; other CV schemes exist & may perform better for your data (see e.g. scikit-learn docs)
- Cross-validation is the go-to method for model evaluation in **machine learning**, as statistics of the models are often not known in the classical sense.
- Again: caveats about selection bias and independence in data.

Summary: Four Recipes for Hacking Statistics

1. Direct Simulation ✓
2. Shuffling ✓
3. Bootstrapping ✓
4. Cross Validation ✓



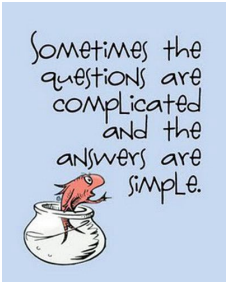
Summary

Sampling Methods allow you to use intuitive **computational** approaches in place of often non-intuitive statistical rules.

If you can write a for-loop you can do statistical analysis.

Summary

Sampling Methods allow you to use intuitive **computational** approaches in place of often non-intuitive statistical rules.
If you can write a for-loop you can do statistical analysis.



– Dr. Seuss (attr)

Quiz/Poll

Why can we use simulation-based methods to compare distributions?

- (A) because the data always follows a Gaussian distribution
- (B) because of the central limit theorem
- (C) because computers generate truly random numbers
- (D) because the sample size is always large enough to avoid any error

Refs used to build this lecture

- Zico Kolter's course on Data Science.
- https://speakerdeck.com/jakevdp/statistics-for-hackers
- https://www.huber.embl.de/msmb/Chap-Testing.html#multiple-testing
- If you want to read more: ISLR book Section 5, slides from previous years